

BIG SMOKE - TOXIC SMOKE DETECTION AND ODOR NEUTRALIZATION ROBOT

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ABSTRACT

Currently, the rate of e-cigarette smoking among Vietnamese students is increasing rapidly, seriously affecting the health of those around them. According to a 2019 survey by the World Health Organization, the rate of e-cigarette use among 15-17-year-old students in Vietnam is 2.6%. By 2022, another survey showed that this rate among 13-15-year-old students had increased to 3.5%. This shows that within 3 years, the rate of e-cigarette use among young people has increased rapidly.

In particular, cigarette smoke contains many gases that are harmful to human health, such as Butadiene, Arsenic, Benzene, Cadmium, and Chromium. Inhaling large amounts of smoke over a long period of time can cause respiratory infections and increase the severity of respiratory diseases.

In addition to cigarette smoke, some other gases, such as CO, generated from fires, internal combustion engines, and burning coal and wood in unventilated places, can also cause asphyxiation, leading to death. Flammable gases leaking into the environment also have a very high risk of causing fire and explosion.

INTRODUCTION

Currently, schools have many solutions to reduce the number of students using tobacco. However, those solutions only stop at propaganda and mobilizing students through propaganda campaigns about the harmful effects of tobacco. At schools, there are patrols for students smoking in the hallways, but this is not significant because the school area is too large and there is not enough human resources.

In addition, there are air purifiers on the market to remove cigarette smoke, but they are limited to a certain area. And if you want to eliminate the smell of cigarettes, you must combine an additional essential oil sprayer, which is very cumbersome for the user.

With the 4.0 industrial revolution, the application of IoT and AI technology to production and human life has become popular. Therefore, I decided to develop the "Big Smoke" project to combine and improve functions to help improve the effectiveness of detecting, filtering, and deodorizing cigarettes in the school

environment. At the same time, the project also has the ability to detect and early warn of fire and explosion cases, helping people promptly find solutions.

I will design a robot that integrates individual features of existing devices on the market. This robot has the ability to move around to detect places with cigarette smoke, then filter the air in place and spray deodorizing essential oils.

METHOD

Step 1: Literature review

During the implementation of this project, I studied various documents to gain knowledge and gather information, forming the scientific and practical basis for the topic, as follows:

- Understanding the harmful effects of tobacco on the human body.
- Studying the composition of toxic smoke.
- Researching the principles of operation and functionalities of the following hardware components:
 - Arduino Mega
 - ESP32 Board
 - GPS Module
 - Air quality sensors: MQ135, MQ2, MQ136, DHT11, ZH03B, MQ7
 - Infrared Camera AMG8833
 - OV7670T Camera
 - Exhaust fan
 - Touchscreen display
 - Memory card/USB or hard drive
 - DC encoder motor
 - UV light
 - Filtration devices (negative ion generator + filters)
 - Essential oil diffuser
 - Lidar sensor
 - Motor driver circuit L298n
 - Raspberry Pi
- Learning how to build and manage a Web Server.

- Exploring front-end and back-end design approaches for Web Servers.
- Understanding the structure, principles of operation, and connection of GPS modules to the control circuit.
- Researching documentation on li-ion battery systems and charging circuits to provide power to the device.
- Designing a web platform to assist teachers and students in monitoring air quality in specific areas and tracking the frequency of toxic smoke occurrences.
- Exploring the features and applications of the NodeMCU module and air quality sensors.
- Learning how to design 3D models for creating the product's frame.
- Studying existing products with similar functionalities available on the market and comparing them.

Product	Example	Mobility	Ability to filter air and deodorize	Ability to detect and warn of fire	Ability to save air quality information
Big Smoke		x	x	x	x
Air purifier Xiaomi Smart Air Purifier 4 compact EU (BHR5860EU) 27W			x		
Fire alarm Horing NQ-418				x	

Table 1: Compare Big Smoke products with devices on the market

Conclusion: It can be seen that the products on the market have one or several features; however, no product has integrated all the features into one device that can be suitable for the conditions of schools in Vietnam. From there, the AirGuard team built the "Big Smoke" product model to solve the remaining problems.

Step 2: Idea

Using IoT (Internet of Things) technology, the product can locate and monitor areas where toxic smoke frequently appears. The web server can record these areas and provide air quality alerts within the school, enabling the administration to identify zones where students smoke and warn about harmful air quality. This helps provide information about smoking-prone areas and alerts students about hazardous air conditions.

The device includes the following functionalities, implemented by specific hardware components:

- + Air Composition Analysis and Smoke Detection

- Sensors (MQ136, MQ135, MQ2, DHT11, and ZH03B): Measure air quality and detect harmful components such as toxic smoke and fine dust particles.

- + Smoke Extraction, Filtration, Disinfection, and Deodorization

- Exhaust Fan: Draws toxic smoke into the system.
 - Air Filter: Purifies the smoke using internal filtration.
 - Essential Oil Diffuser: Releases essential oils to neutralize odors in the surrounding area.

- + Data Storage and Localization

- GPS Module: Logs locations where toxic smoke frequently appears.
 - ESP32 Board: Tracks and patrols these areas by sending data to the web server.
 - Lidar Sensor: Measures distances and detects obstacles around the device.

- + Real-Time Monitoring via Camera

- OV7670 Camera: Captures images for monitoring.
 - Web Accessibility: Provides live camera feeds through the web server.

- + Fire Detection and Alerts

- Infrared Camera (AMG8833): Detects abnormally high temperatures to identify potential fire hazards.

- ESP32 Board: Sends immediate alerts to the web server for safety measures.

- MQ2 and MQ7 Sensors: Detect flammable gases and toxic substances.

- SMS Alerts: Sends text notifications to the administrator.

+ Nearby Area Sanitization

- UV Light: Disinfects the surrounding area

Products are designed to serve key audiences

1. For schools:

- The product assists teachers in identifying areas where students may smoke and places with poor air quality, enabling them to apply appropriate measures to improve the learning environment.

2. For students:

- The product provides warning information about areas with poor air, helping students recognize and stay away from areas with air harmful to their health.

3. For families:

- The product helps family members monitor indoor air quality, and detect toxic smoke and other pollutants to provide timely warnings. This helps protect the health of everyone, especially children and the elderly, who are vulnerable to poor air quality.
- In addition, the product can detect signs of fire and explosion and send an immediate warning so the family can handle it promptly, ensuring safety for the whole house.

Step 3: Technology solutions

To realize ideas into actual product models, proposed technological solutions include 3D design models of circuits and products, general block diagrams of the system, flow charts, mathematics, and operating principles of the device.

- **3D design model for product:**

- Material:
 - Frame: shaped aluminum
 - Formex
 - Plastic
- Specifications:
 - Length: 300mm
 - Width: 300mm
 - Height: 250mm
 - Wheel diameter: 90mm

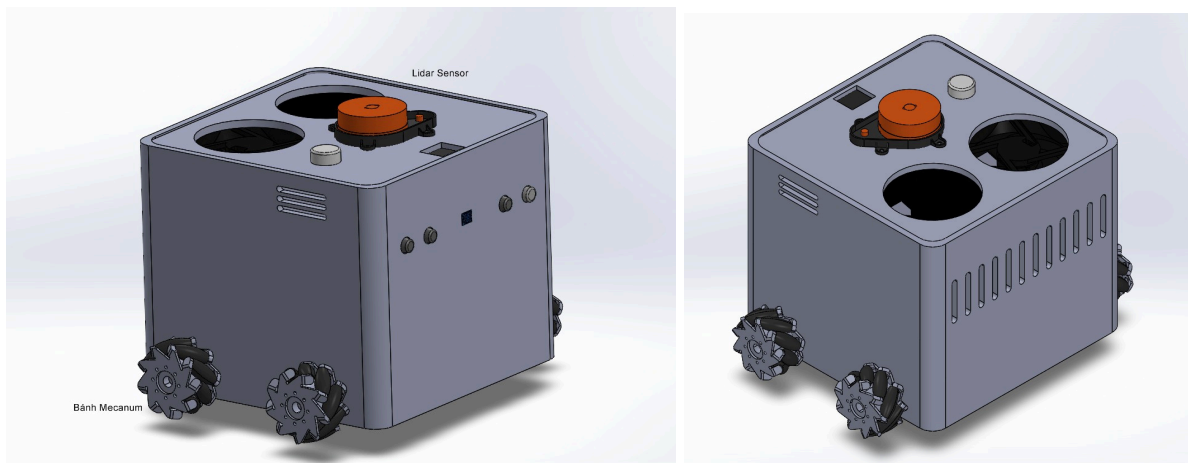


Figure 1: General model

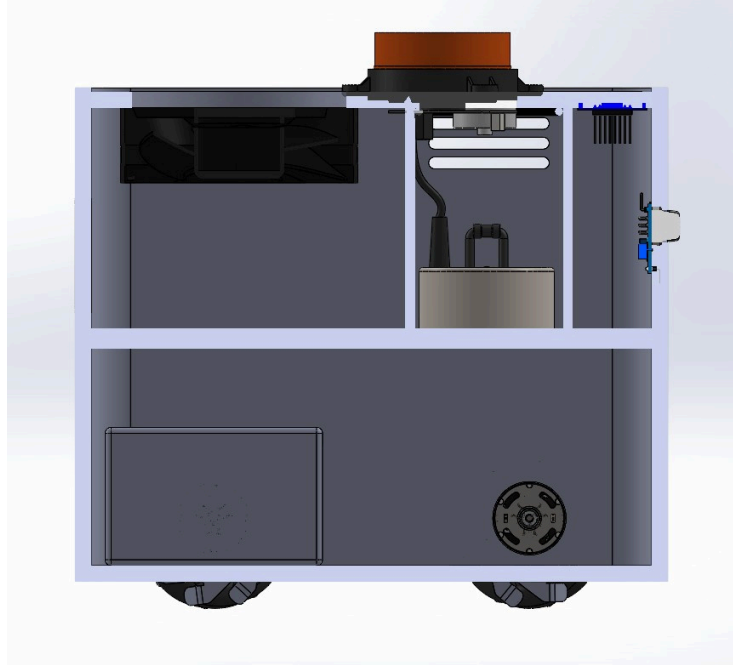


Figure 2: Longitudinal section of the robot

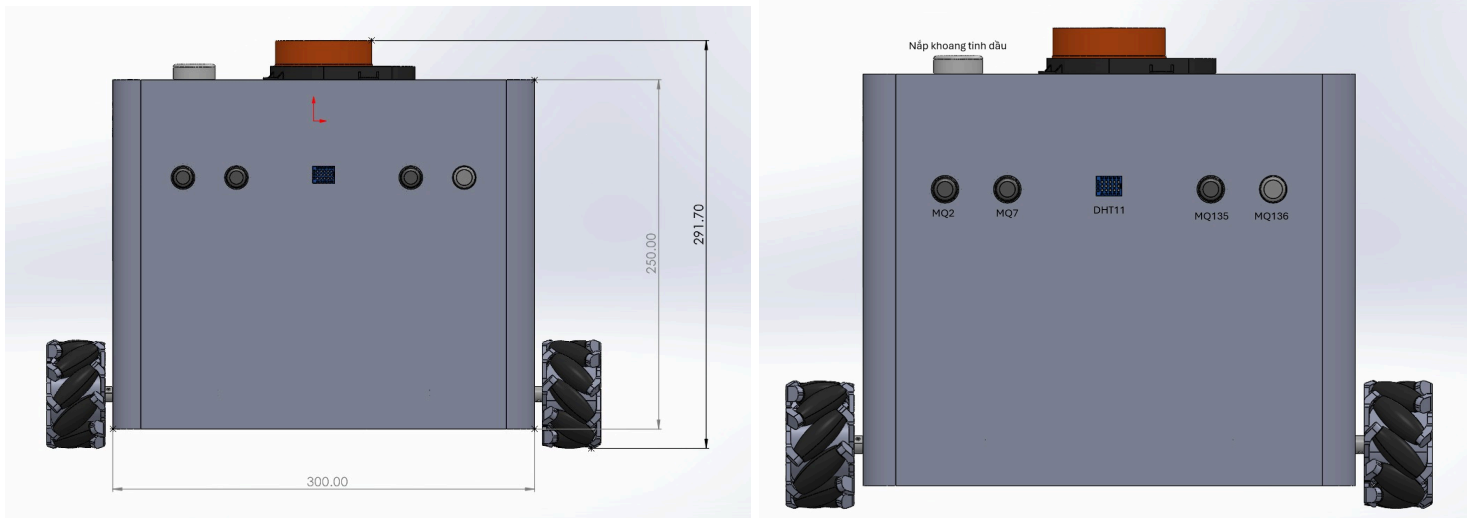


Figure 3: Front view of the robot

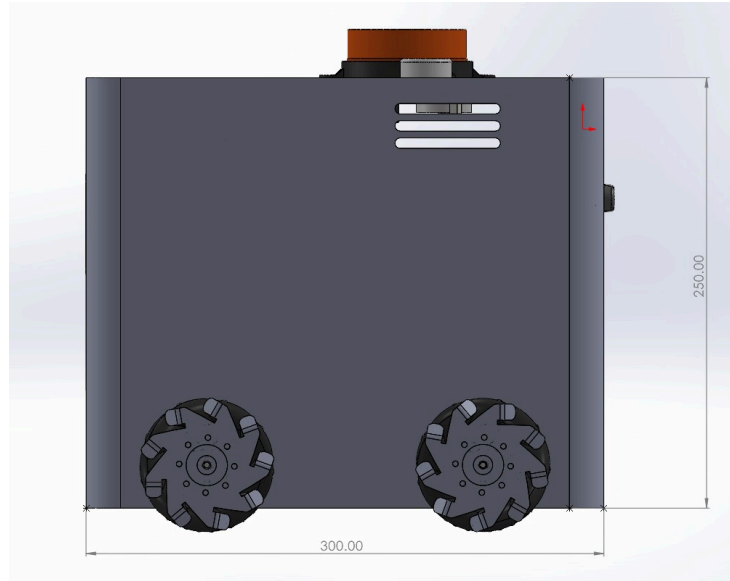


Figure 4: Robot side view

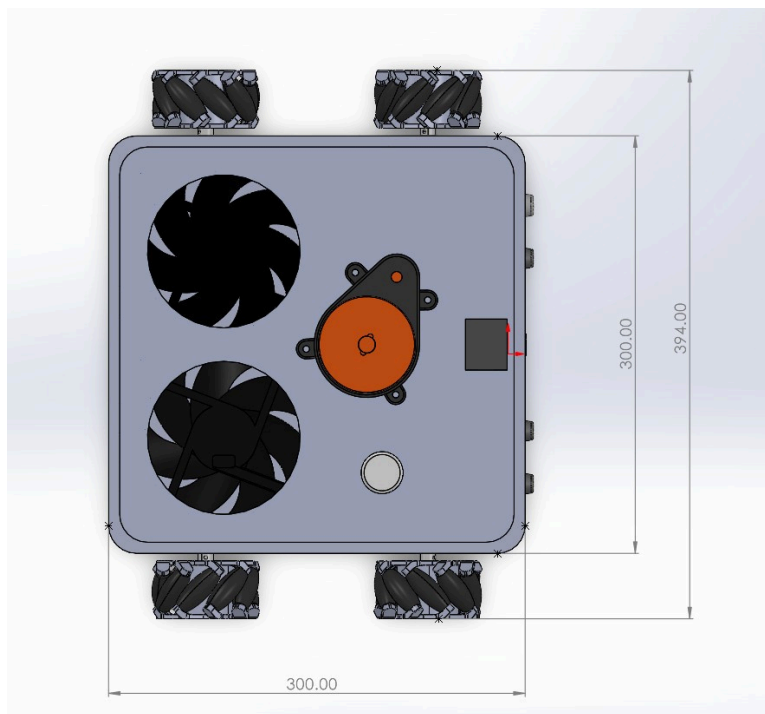


Figure 5: Robot lid

- *3D circuit simulation of the device:*

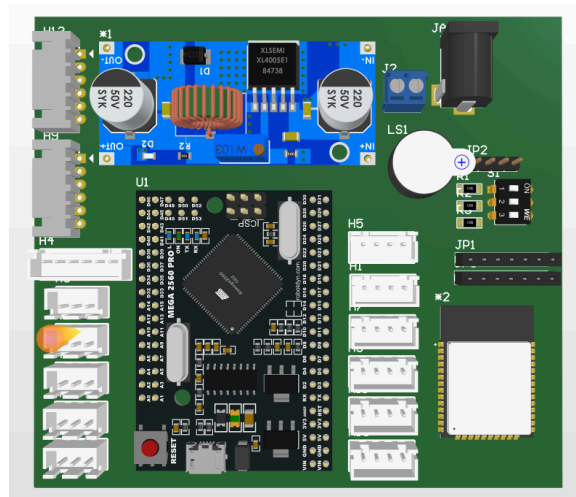


Figure 6: 3D circuit simulation of the device

- *System block diagram of the device:*

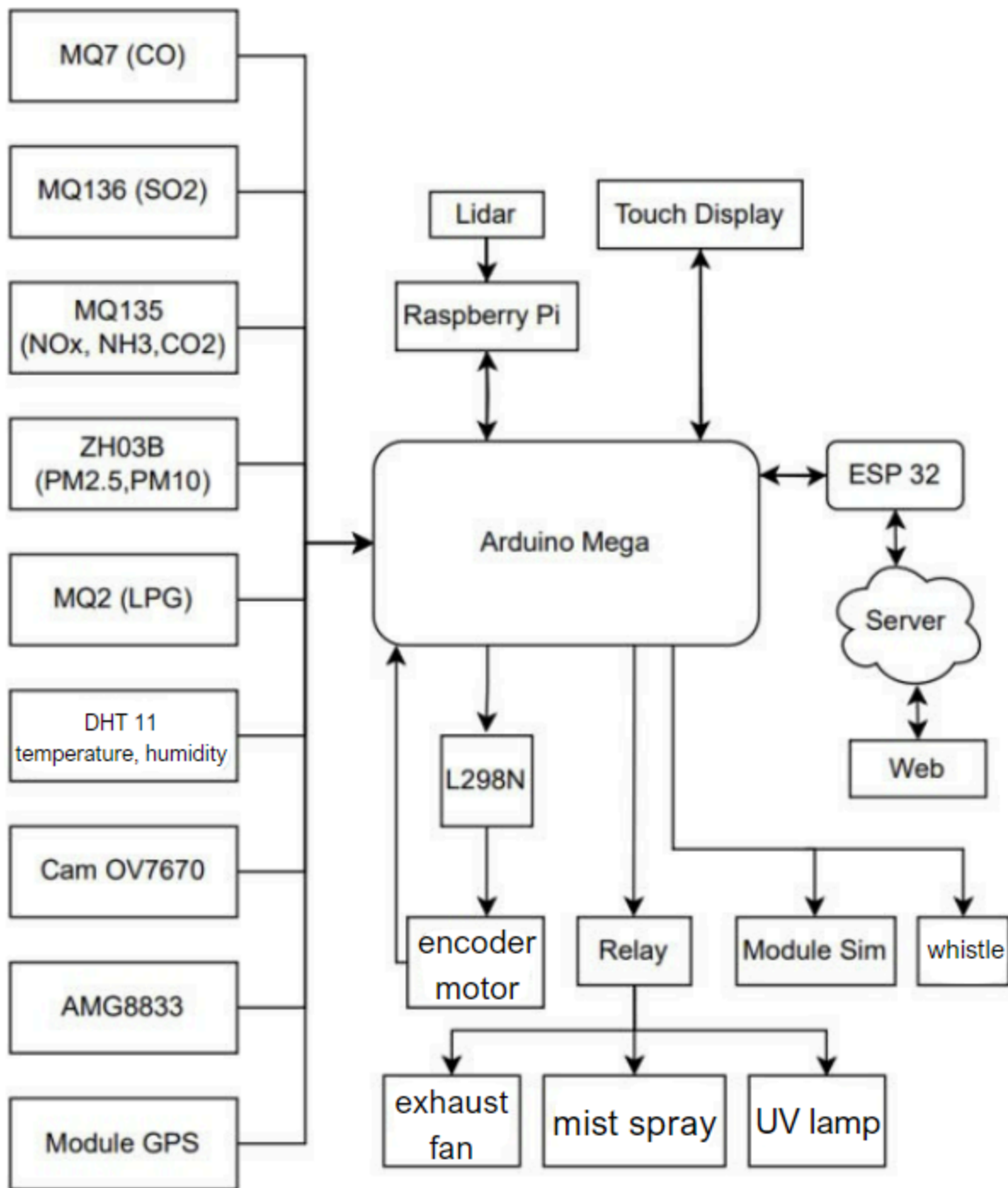


Figure 7: System block diagram of the device

- Flow chart of the moving part algorithm of the device:

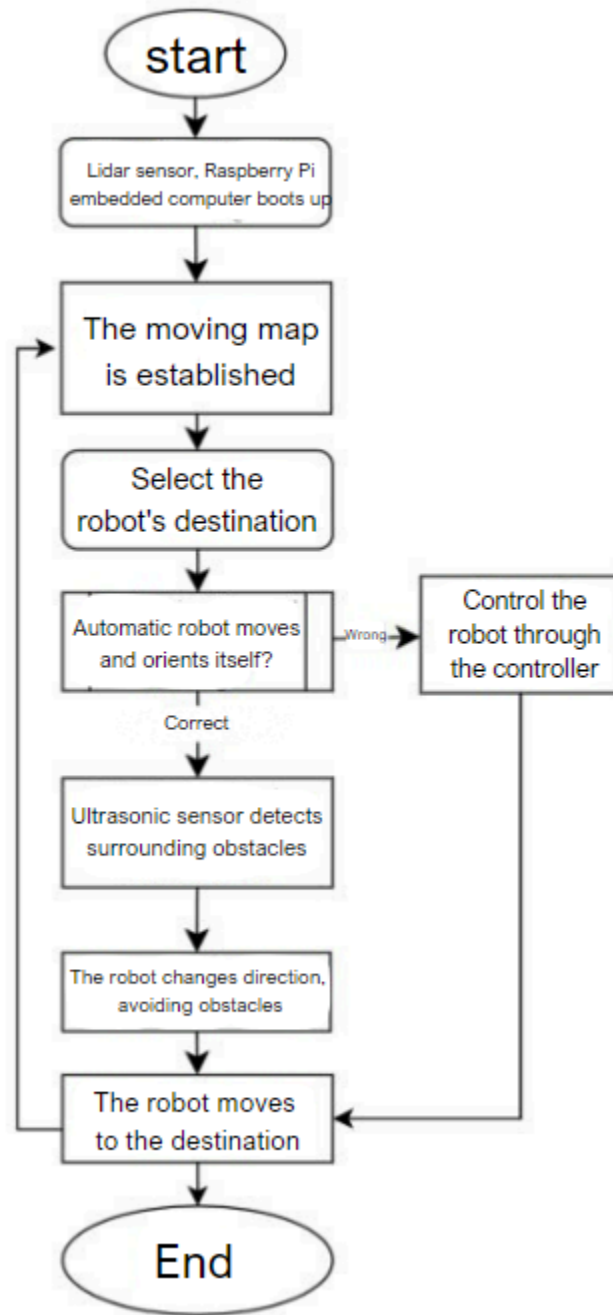


Figure 8: Flow chart of the moving part algorithm of the device

- Flow chart of the device's control algorithm:

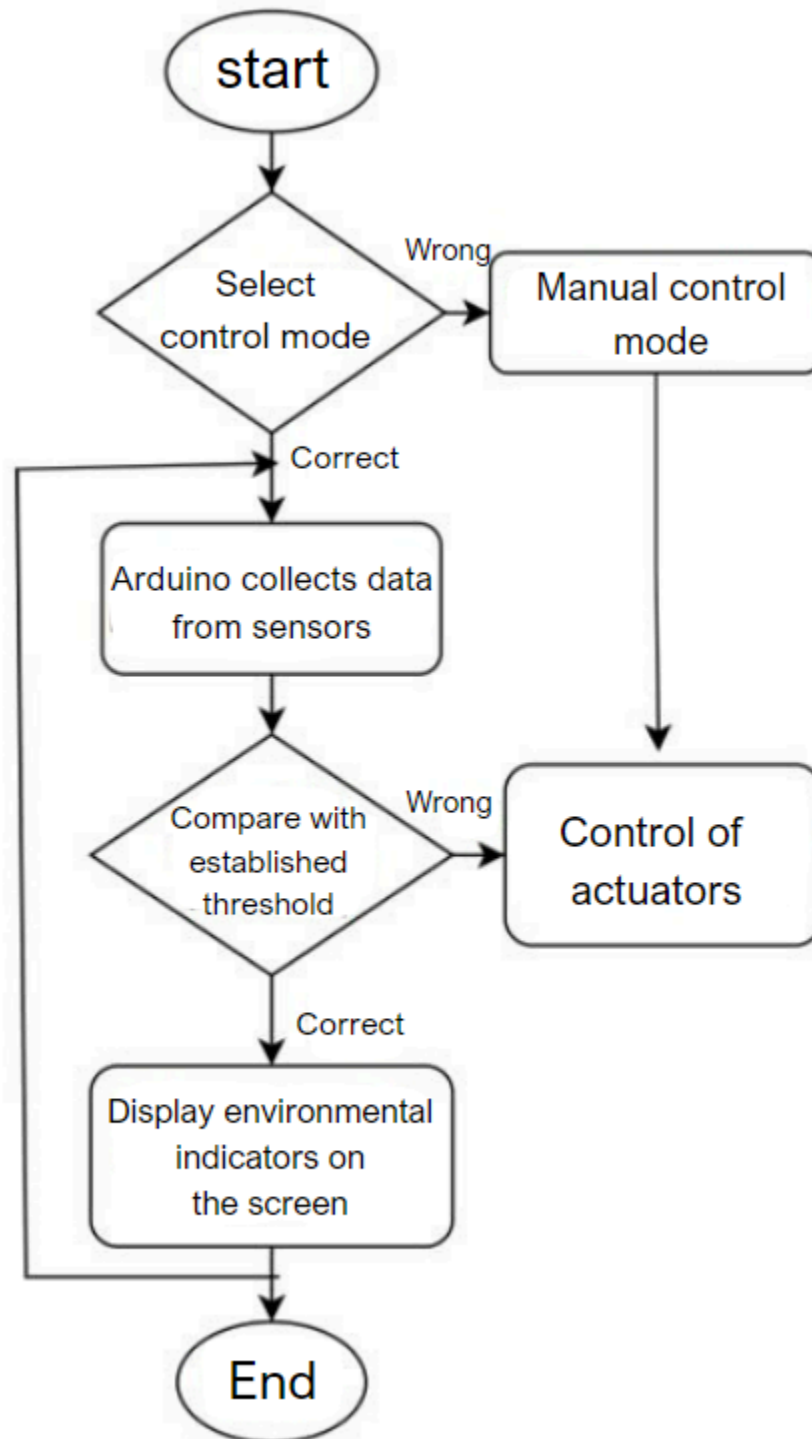


Figure 9: Flow chart of the device's control algorithm

- **Operating principle of the device:**

+ Measure, warn, handle:

- When powering the system, the default running mode is Auto mode.
 - + In Auto mode, after the system is powered and booted, the Arduino board will read the values of the sensors. The values after being read from the sensor will be sent by the Arduino Nano board to the NodeMCU module to be sent to the server and displayed on the web.
 - + Conditions for controlling the exhaust fan and mist sprayer to operate automatically. Specifically :
 - When detecting toxic smoke or dust concentration exceeding the safe threshold ($>45 \mu\text{g}/\text{m}^3$), the machine will start the exhaust fan to filter the air, turning off when the dust concentration is within the safe threshold ($<30 \mu\text{g}/\text{m}^3$)
 - When the CO concentration exceeds the safety threshold ($>30 \text{ ppm}/\text{m}^3$), the machine will automatically warn, and turn on the air filtering function, the CO gas will be filtered through the activated carbon membrane, otherwise, the CO concentration in the air will be low. The machine will stop filtering air within the safety threshold ($< 10 \text{ ppm}/\text{m}^3$).
 - Gas concentration ($> 15\%$) the machine will automatically warn.
- Gas concentration ($> 15\%$) the machine will automatically warn. In addition, during operation, the system can be switched to Manual mode via the push of a button: In this mode, the system reads the sensor, displays it on the screen, and sends it to NodeMCU for transmission to the website. The air filters and mist sprayers do not operate automatically but are turned on via a push-button system during a preset time.

+ Movement:

- Use the server as a laptop to process the signal.
- Raspberry Pi acts as an intermediary device connecting the robot's moving parts to the server, which is a laptop connected to the Raspberry Pi via Raspi's wifi using the SSH command to remotely control the Raspberry Pi.

- Raspberry Pi communicates 2-way with the Arduino mega board via uart to transmit velocity commands from the server to the Arduino and the Arduino will process the signal and output appropriate voltage levels to the h-bridge to control the motor. The motor responds to the signal from the encoder signal pin to the Arduino and sends it to the Raspberry for transmission to the server to calculate and determine the robot's position on the map.
- The lidar sensor communicates Uart with Raspberry to transmit RAW signals (angle and distance) to the server via the embedded Raspberry computer and releases signals to determine the direction and position of the robot. Most of the processing steps are on the server, which is a laptop because the Raspberry Pi is quite weak.

FEASIBILITY

The product has characteristics that prove its feasibility:

- The product is compact and easy to move across terrain thanks to the use of Mecanum wheels.
- Using components and materials that are easy to find and cheap, can optimize costs when producing in large quantities.
- Short manufacturing and assembly time, can easily transfer technology and mass produce.

Disadvantages:

- The hardware design is not yet optimized.
- The product has not been tested on a large scale to evaluate quality.

Estimated price:

Number	Name	Amount	Price	Example
1	Arduino Mega	1	150.000VND	

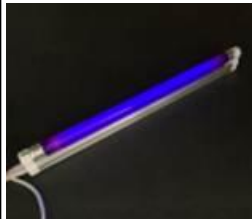
2	MQ135	1	30.000VND	
3	MQ136	1	100.000VND	
4	MQ7	1	30.000VND	
5	MQ2	1	30.000VND	
6	DHT11	1	35.000VND	

7	AMG8833	1	300.000VND	
8	ZH03B	1	250.000VND	
9	LCD TFT	1	155.000VND	
10	Module GPS NEO-6M	1	110.000VND	
11	Negative ion technology filter membrane	2	90.000VND	

12	Filter HEPA	1	100.000VND	
13	Exhaust fan	2	100.000VND	
14	Wheels mecanum	4	100.000VND	
15	Mist spray head	1	80.000VND	

16	Sensor Lidar	1	1.000.000VND	
17	PIN 12V	1	200.000VND	
18	Brush	1	55.000VND	
19	Camera OV7670	1	50.000VND	

20	Electronic Motor DC JGB37-545	4	500.000 VND	
21	Raspberry Pi	1	1.500.000VND	
22	3D printing materials	2 cuộn	400.000VND	 0
23	Motor control circuit L298n	2	40.000VND	
24	Relay	1	35.000VND	

25	UVLight	1	50.000VND	
26	Rent a domain name	1 year	300.000VND	
27	Inspection, measurement and testing, costs incurred		1.000.000VND	
Total			6.800.000VND	

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